

WIEN2k 24.1 Install with OneAPI (ifx)

August 22, 2026

Linux Operating System Version (lsb_release -a): Rocky Linux release 8.10 (Green Obsidian)
Fortran Compiler Version (ifx -v): ifx version 2026.1.1
WIEN2k Version (cat \$WIENROOT/WIEN2k_VERSION): 24.1 (Release 1/8/2024)

The Rocky Linux 8.10 was already installed, but if that is needed it should be available at:

<https://rockylinux.org/download>

Operating system packages needed were installed before proceeding with step 1 given below. If that hasn't been done, then install them:

```
[username@localhost ~]$ sudo dnf upgrade
...
[username@localhost ~]$ sudo dnf install redhat-lsb-core tcsh make cmake autoconf libtool
...
[username@localhost ~]$ sudo dnf install gnuplot gedit patch mesa-libGL-devel
...
[username@localhost ~]$ sudo dnf install mesa-libGLU-devel tcl-devel tk-devel
...
[username@localhost ~]$ sudo dnf install fftw-devel libXmu-devel perl gcc-gfortran
...
[username@localhost ~]$ sudo dnf install python3.12-pip
...
[username@localhost ~]$ python3.12 -m pip install --user pytest
...
[username@localhost ~]$ python3.12 -m pip install --user numpy
...
```

Note: If prompted with “Is this ok [y/N]”, enter: y.

Togl 2.0 was installed following

<https://mailman.ijs.si/pipermail/xcrysden/2024-September/002152.html> :

```
[username@localhost ~]$ wget https://sourceforge.net/projects/togl/files/Togl/2.0/Togl2.0-
src.tar.gz/download
...
[username@localhost ~]$ tar xvf download
...
[username@localhost ~]$ cd Togl2.0/
[username@localhost Togl2.0]$ ./configure --with-tcl=/usr/lib64 --with-tk=/usr/lib64 --
enable-64bit
...
```

```
[username@localhost Togl2.0]$ wget https://raw.githubusercontent.com/gsabo/xcrysdn-Patches/main/togl.patch
```

...

```
[username@localhost Togl2.0]$ patch -b togl.c togl.patch
```

```
[username@localhost Togl2.0]$ make
```

```
[username@localhost Togl2.0]$ ln -s libTogl2.0.so libTogl.so.2
```

The XCrySDen 1.6.2 was installed using:

```
[username@localhost Togl2.0]$ cd ..
```

```
[username@localhost ~]$ wget http://www.xcrysden.org/download/xcrysden-1.6.2-linux_x86_64-shared.tar.gz
```

...

```
[username@localhost ~]$ tar xvf xcrysden-1.6.2-linux_x86_64-shared.tar.gz
```

...

```
[username@localhost ~]$ gedit ~/.bashrc
```

...

```
export XCRYSDEN_TOPDIR=$HOME/xcrysden-1.6.2-bin-shared
```

```
export PATH=$PATH:$XCRYSDEN_TOPDIR
```

```
export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:$HOME/Togl2.0
```

```
[username@localhost ~]$ source ~/.bashrc
```

Of note, for XCrySDen to work with Wayland, toglOpt(accum) needs set to false in the /home/username/xcrysden-1.6.2-bin-shared/Tcl/custom-definitions file:

<https://www.mail-archive.com/wien@zeus.theochem.tuwien.ac.at/msg23440.html>

Installed OneAPI [<https://www.intel.com/content/www/us/en/docs/oneapi-toolkit/installation-guide-linux/latest/install-oneapi-toolkit-with-yum-dnf.html#HPC-YUM-DNF>] using:

```
[username@localhost ~]$ tee > /tmp/oneAPI.repo << EOF
```

```
[oneAPI]
```

```
name=Intel® oneAPI repository
```

```
baseurl=https://yum.repos.intel.com/oneapi
```

```
enabled=1
```

```
gpgcheck=1
```

```
repo_gpgcheck=1
```

```
gpgkey=https://yum.repos.intel.com/intel-gpg-keys/GPG-PUB-KEY-INTEL-SW-PRODUCTS.PUB
```

```
EOF
```

...

```

[username@localhost ~]$ sudo mv /tmp/oneAPI.repo /etc/yum.repos.d
[username@localhost ~]$ sudo dnf install intel-hpckit
...
Is this ok [y/N]: y

...

Is this ok [y/N]: y
...

Is this ok [y/N]: y
...

[username@localhost ~]$ sudo dnf install intel-oneapi-toolkit

...
[username@localhost ~]$ echo "source /opt/intel/oneapi/setvars.sh --force" >> ~/.bashrc
[username@localhost ~]$ grep oneapi ~/.bashrc
source /opt/intel/oneapi/setvars.sh --force
[username@localhost ~]$ source ~/.bashrc

:: initializing oneAPI environment ...
  bash: BASH_VERSION = 4.4.20(1)-release
  args: Using "$@" for setvars.sh arguments: --force
:: advisor -- latest
:: ccl -- latest
:: compiler -- latest
:: dal -- latest
:: debugger -- latest
:: dev-utilities -- latest
:: dnnl -- latest
:: dpcpp-ct -- latest
:: dpl -- latest
:: ipp -- latest
:: ippcp -- latest
:: ishmem -- latest
:: mkl -- latest
:: mpi -- latest
:: tbb -- latest
:: tcm -- latest
:: umf -- latest
:: vtune -- latest
:: oneAPI environment initialized ::
[username@localhost ~]$ ifx -v

```

```

ifx version 2026.1.1
[username@localhost ~]$ icx -v
Intel(R) oneAPI DPC++/C++ Compiler 2026.1.1 (2026.1.1.20260724)
...
[username@localhost ~]$ mpiifx -v
...
ifx version 2026.1.1

```

1. The following was entered into the terminal to install libxc:

```

[username@localhost ~]$ cd ~
[username@localhost ~]$ wget https://gitlab.com/libxc/libxc/-/archive/7.1.2/libxc-7.1.2.tar.bz2
...
[username@localhost ~]$ tar xvf libxc-7.1.2.tar.bz2
...
[username@localhost ~]$ cd libxc-7.1.2/
[username@localhost libxc-7.1.2]$ cmake -H. -Bobjdir
-DCMAKE_INSTALL_PREFIX=$HOME/libxc-7.1.2/ -DBUILD_SHARED_LIBS=false
-DENABLE_FORTRAN=ON -DCMAKE_C_COMPILER=icx
...
[username@localhost libxc-7.1.2]$ cd objdir/
[username@localhost objdir]$ make
...
[username@localhost objdir]$ ctest
Test project /home/username/libxc-7.1.2/objdir
  Start 1: special-functions
1/3 Test #1: special-functions ..... Passed   0.00 sec
  Start 2: api
2/3 Test #2: api ..... Passed   0.00 sec
  Start 3: distinct
3/3 Test #3: distinct ..... Passed   0.01 sec

100% tests passed, 0 tests failed out of 3

Total Test time (real) =  0.02 sec
[username@localhost objdir]$ make install
...
[username@localhost objdir]$ ls ~/libxc-7.1.2/lib64
cmake libxc.a libxcf03.a pkgconfig

```

2. The following was entered into the terminal to install fftw:

```

[username@localhost objdir]$ cd ~
[username@localhost ~]$ wget https://www.fftw.org/fftw-3.3.11.tar.gz
...

```

```
[username@localhost ~]$ tar xvf fftw-3.3.11.tar.gz
...
[username@localhost ~]$ cd fftw-3.3.11/
[username@localhost fftw-3.3.11]$ ./configure FCC=ifx CC=icx MPICC=mpiicx --enable-mpi
--prefix=$HOME/fftw-3.3.11
...
[username@localhost fftw-3.3.11]$ make
...
[username@localhost fftw-3.3.11]$ make install
...

[username@localhost fftw-3.3.11]$ ls ~/fftw-3.3.11/include ~/fftw-3.3.11/lib
/home/username/fftw-3.3.11/include:
fftw3.f  fftw3.h  fftw3l-mpi.f03  fftw3-mpi.h
fftw3.f03  fftw3l.f03  fftw3-mpi.f03  fftw3q.f03

/home/username/fftw-3.3.11/lib:
cmake  libfftw3.a  libfftw3.la  libfftw3_mpi.a  libfftw3_mpi.la  pkgconfig
```

3. The following was entered into the terminal to install ELPA:

```
[username@localhost fftw-3.3.11]$ cd ~
[username@localhost ~]$ wget
https://elpa.mpcdf.mpg.de/software/tarball-archive/Releases/2026.02.002/
elpa-2026.02.002.tar.gz
...
[username@localhost ~]$ tar xvf elpa-2026.02.002.tar.gz
...
[username@localhost ~]$ cd elpa-2026.02.002/
...

[username@localhost elpa-2026.02.002]$ ./configure FC=mpiifx CC=mpiicx
SCALAPACK_LDFLAGS="-L$MKLROOT/lib/intel64 -lmkl_scalapack_lp64 -lmkl_intel_lp64
-lmkl_sequential -lmkl_core -lmkl_blacs_intelmpi_lp64 -lpthread -lm -ldl -liomp5 -lm -Wl,-
rpath,$MKLROOT/lib/intel64" SCALAPACK_FC_FLAGS="-L$MKLROOT/lib/intel64
-lmkl_scalapack_lp64 -lmkl_intel_lp64 -lmkl_sequential -lmkl_core -lmkl_blacs_intelmpi_lp64
-lpthread -lm -I$MKLROOT/include/intel64/lp64" --prefix=$HOME/elpa-2026.02.002
FCFLAGS=-O3 CFLAGS="-O3 -mfma -funsafe-math-optimizations -ftree-vectorize" LIBS="-
lmpi -lmpifort" --disable-avx512
...

```

The following ELPA2 kernels will be build:

- real_generic
- real_generic_simple
- real_generic_simple_block4
- real_generic_simple_block6
- real_sse_block2
- real_sse_block4
- real_sse_block6
- real_sse_assembly
- real_avx_block2
- real_avx_block4
- real_avx_block6
- real_avx2_block2 (default)
- real_avx2_block4
- real_avx2_block6
- complex_generic
- complex_generic_simple
- complex_sse_block1
- complex_sse_block2
- complex_sse_assembly
- complex_avx_block1
- complex_avx_block2
- complex_avx2_block1 (default)
- complex_avx2_block2

This version of ELPA support the minimal API version: 20170403

The current API version is: 20260202

This version of ELPA support the minimal autotuning version: 20171201

The current autotune version is: 20260202

...

```
[username@localhost elpa-2026.02.002]$ make
```

...

```
[username@localhost elpa-2026.02.002]$ make install
```

...

```
[username@localhost elpa-2026.02.002]$ ls ~/elpa-2026.02.002/lib ~/elpa-2026.02.002/elpa/  
/home/username/elpa-2026.02.002/elpa/:
```

```

elpa_build_config.h      elpa_explicit_name.c  elpa_generic.h
elpa_build_config.h.in   elpa_explicit_name.h  elpa.h
elpa_configured_options.h elpa_explicit_name.lo elpa_simd_constants.h
elpa_configured_options.h.in elpa_explicit_name.o  elpa_version.h
elpa_constants.h         elpa_generated_c_api.h elpa_version.h.in
elpa_constants.h.in      elpa_generated.h

```

```

/home/username/elpa-2026.02.002/lib:
libelpa.a libelpa.so libelpa.so.19.5.2
libelpa.la libelpa.so.19 pkgconfig

```

4. Go to:
http://www.wien2k.at/reg_user/index.html
5. Click "Code download (after registration)"
6. Enter your username and password that you were given when you purchased WIEN2k.
7. Click the link "here" in "You can download the complete source code (about 170 MB) from here" to download "WIEN2k_24.1.tar". You should now have the source code package for WIEN2k 24.1.
8. If you have the folder "WIEN2k" in your Linux operating system home directory (for example at /home/username/WIEN2k, where username is replaced by your user name). You can rename it something else, such as "WIEN2k23.2".
9. The following was entered into the terminal to install WIEN2k 24.1:

```

[username@localhost elpa-2026.02.002]$ cd ~
[username@localhost ~]$ mkdir WIEN2k
[username@localhost ~]$ cd WIEN2k
[username@localhost WIEN2k]$ ls
WIEN2k_24.1.tar
[username@localhost WIEN2k]$ tar xvf WIEN2k_24.1.tar
...
[username@localhost WIEN2k]$ gunzip *.gz
[username@localhost WIEN2k]$ chmod +x ./expand_lapw

[username@localhost WIEN2k]$ export LC_NUMERIC=en_US.UTF-8

[username@localhost WIEN2k]$ ./expand_lapw
...
continue (y/n)
y
...

```

Applied patches at: <https://github.com/gsabowien2k-patches/tree/master/24.1>

```
[username@localhost WIEN2k]$ ./siteconfig
```

```
...
```

```
continue or stop (c/s) c
```

```
    Press RETURN to continue
```

```
...
```

```
    Selection: LI
```

```
...
```

```
    Press RETURN to continue
```

```
...
```

```
    Your compiler: ifx
```

```
...
```

```
    Your compiler: icx
```

```
...
```

```
    Press RETURN to continue
```

```
...
```

```
Hit Enter to continue
```

```
...
```

```
Would you like to use LIBXC (needed ONLY for self-consistent gKS mGGA calculations, for the stress tensor and experts who want to play with different DFT options. It must have been installed before)? (y,N):
```

```
y
```

```
...
```

```
Do you want to automatically search for LIBXC installations? (Y,n):
```

```
Y
```

```
Please specify a comma separated list of directories to search! (If no list is entered, /usr/lib64, /usr/local and /opt will be searched as default):  
/home/username/libxc-7.1.2
```

```
...
```

```
More than one libxc was found in the specified directory(ies).
```

```
Pick one of the following (enter the number of the line of your choice or 0 to manually specify a path)!:
```

```
/home/username/libxc-7.1.2/objdir/libxc.a
```

```
/home/username/libxc-7.1.2/lib64/libxc.a
```

```
2
```

```
...
```

```
    Press RETURN to continue
```

```
...
```

```
Do you want to automatically search for FFTW installations? (Y,n):
```

```
Y
```

```
Please specify a comma separated list of directories to search! (If no list is entered, /usr/lib64, /usr/local and /opt will be searched as default):
```



```

/home/username/fftw-3.3.11
...
Do you want to use a FFTW version from the list above? (Y,n):
Y
Please enter the line number of the chosen version or enter 0 to manually specify your choice!:
2
...
Please specify the target achitecture of your FFTW library (e.g. lib64) or accept present choice
(enter): lib
...
Please specify the name of your FFTW library or accept present choice (enter): fftw3
...
Is this correct? (Y,n): Y
...
Current settings:
M OpenMP switch:      -qopenmp
O Compiler options:   -O -FR -mp1 -w -prec_div -pc80 -pad -ip -DINTEL_VML
-traceback -assume buffered_io -I$(MKLROOT)/include
L Linker Flags:       $(FOPT) -L$(MKLROOT)/lib/$(MKL_TARGET_ARCH) -lpthread
-lm -ldl -liomp5
P Preprocessor flags  '-DParallel'
R R_LIBS (LAPACK+BLAS): -lmkl_intel_lp64 -lmkl_intel_thread -lmkl_core
F FFTW options:       -DFFTW3 -I/home/username/fftw-3.3.11/include
FFTW-LIBS:            -L/home/username/fftw-3.3.11/lib -lfftw3
X LIBX options:       -DLIBXC -I/home/username/libxc-7.1.2/include
LIBXC-LIBS:           -L/home/username/libxc-7.1.2/lib64 -lxcf03 -lxc

S Save and Quit
...
Selection: S
...
Press RETURN to continue
...
Shared Memory Architecture? (y/N):y
Do you know/need a command to bind your jobs to specific nodes?
(like taskset -c). Enter N / your_specific_command: N
...
Do you have MPI, ScaLAPACK, ELPA, or MPI-parallel FFTW installed and intend
to run finegrained parallel?
...
(y/N) y
...
Your compiler: mpiifx
...
Do you want to use a present ScaLAPACK installation? (Y,n): Y

```

```

...
Do you want to use the MKL version of ScaLAPACK? (Y,n):Y
Do you use Intel MPI? (Y,n):Y
...
Is this correct? (Y,n): Y
...
    Press RETURN to continue
...
Do you want to use ELPA? (y,N):
y
...
Do you want to automatically search for ELPA installations? (Y,n):
Y
Please specify a comma-separated list of directories to search! (If no list is entered,
/usr/lib64 /usr/local and /opt will be searched as default):
/home/username/elpa-2026.02.002
Finding the required ELPA files in /home/username/elpa-2026.02.002 ....

More than one version of ELPA found:
/home/username/elpa-2026.02.002/lib
/home/username/elpa-2026.02.002/.libs

Pick one (enter line number) or enter 0 to manually specify a path!:
1
Present root directory of ELPA is: /home/username/elpa-2026.02.002/
Please specify the ROOT-path of your ELPA installation (like /usr/local/elpa/) or accept present
path (Enter):
Checking for ELPA version ...
More than one set of include files in your ELPA-ROOT directory. Pick one (enter line number)
or enter 0 to manually specify a version!:
/home/username/elpa-2026.02.002/src/elpa1
/home/username/elpa-2026.02.002/src/elpa2
/home/username/elpa-2026.02.002/src/elpa_generalized
/home/username/elpa-2026.02.002/elpa
/home/username/elpa-2026.02.002/test/Fortran/elpa2
/home/username/elpa-2026.02.002/share/doc/elpa
/home/username/elpa-2026.02.002/.fortran_dependencies/elpa2_print_kernels
/home/username/elpa-2026.02.002/include/elpa-2026.02.002
/home/username/elpa-2026.02.002/include/elpa-2026.02.002/elpa
8
...
Is this correct? (Y,n):Y
...
Please specify the lib-directory of your ELPA installation (e.g. lib or lib64):
lib

```

```

...
Is this correct? (Y,n):Y
...
Please specify the name of your installed ELPA library (e.g. elpa or elpa_openmp):
elpa
Your current lib-directory of your ELPA installation is: lib
Is this correct? (Y,n):Y
The current library name of your ELPA installation is: elpa
Is this correct? (Y,n):Y
...
Is this correct? (Y,n): Y

Press RETURN to continue
...
Please specify your parallel compiler options or accept the recommendations (Enter - default)!:

Please specify your parallel OMP_SWITCH (type "del" to blank it)
or accept the recommendations (Enter - default)!:

Please specify your MPIRUN command or accept the recommendations (Enter - default)!:

Press RETURN to continue
...
Current settings:

Parallel compiler      : mpiifx
SCALAPACK_LIBS         : -lmkl_scalapack_lp64 -lmkl_blacs_intelmpi_lp64
FFTW_PLIBS             : -lfftw3_mpi
ELPA_OPT               : -DELPA
-I/home/username/elpa-2026.02.002/include/elpa-2026.02.002/elpa
-I/home/username/elpa-2026.02.002/include/elpa-2026.02.002/modules
ELPA_LIBS              : -lelpa -L/home/username/elpa-2026.02.002/lib
-Wl,-rpath=/home/username/elpa-2026.02.002/lib
FPOPT(par.comp.options): -O -FR -mp1 -w -prec_div -pc80 -pad -ip -DINTEL_VML
-traceback -assume buffered_io -I$(MKLROOT)/include
OMP_SWITCH             : -qopenmp
MPIRUN command         : mpirun -np _NP_ -machinefile _HOSTS__EXEC_

parallel execution:

RP_LIBS                : $(R_LIBS)
...
Selection: S

```

```

...
Press RETURN to continue
...
Selection: Q
...
Selection: A
...
Compile time errors (if any) were:
                                     <= It should be blank here if successful.
Check file  compile.msg  in the corresponding SRC_* directory for the
compilation log and more info on any compilation problem.

Press RETURN to continue
...
Please enter the full path of the perl program: /usr/bin/perl
...
Press RETURN to continue
...
Please enter the full path to your temporary directory: /tmp
...
Press RETURN to continue
...

```

10. If it is the first time installing WIEN2k on the computer, enter in the terminal:

```

[username@localhost WIEN2k]$ ./userconfig_lapw
...
Specify your preferred editor (default is emacs):
editor shall be: gedit

Set editor to  gedit (Y/n) Y

Specify your preferred DATA directory, where your cases should be
stored (for /home/username/WIEN2k, just enter RETURN key):
DATA directory: /home/username/wiendata

Set DATA directory to  /home/username/wiendata (Y/n) Y
...
Specify your preferred scratch directory, where big case.vector files
can be stored (Recommended is a local directory (maybe /scratch), not a
NFS directory. For your working directory, just enter RETURN key):
scratch directory:

Set scratch directory to  working directory (Y/n) Y

```

Specify your program to read pdf files (default is okular)
(on some Linux systems use xpdf, evince, pdfstudio, ...):evince

...

Set PDFREADER to evince (Y/n) Y

WIEN2k can use OpenMP parallelization on multicore computers.
For details please read the "Parallelization section" of the Usersguide.
Your present computer has 6 cores, but more than 4 (8) cores is useless.
How many cores do you want to use by default (4):1

Set OMP_NUM_THREADS to 1 (Y/n) Y

!!! The following lines will be added to your .bashrc file if you continue !!!
A copy of your current .bashrc will be saved under .bashrc.savelapw !

```
# added by WIEN2k: BEGIN
# -----
alias lsi="ls -aslp *.in*"
alias lso="ls -aslp *.output*"
alias lsd="ls -aslp *.def"
alias lsc="ls -aslp *.clm*"
alias lss="ls -aslp *.scf* /*.scf"
alias lse="ls -aslp *.error"
alias LS="ls -alsp |grep /"
alias pslapw="ps -ef |grep "lapw""
alias cdw="cd /home/username/wiendata"
if [ "$OMP_NUM_THREADS" = "" ]; then export OMP_NUM_THREADS=1; fi
#export LD_LIBRARY_PATH=.....
export EDITOR="gedit"
export SCRATCH=./
if [ "$WIENROOT" = "" ]; then export WIENROOT=/home/username/WIEN2k; fi
export W2WEB_CASE_BASEDIR=/home/username/wiendata
export STRUCTEDIT_PATH=$WIENROOT/SRC_structeditor/bin
export PDFREADER=evince
export PATH=$WIENROOT:$STRUCTEDIT_PATH:$WIENROOT/SRC_IRelast/script-
elastic:$PATH:.
export OCTAVE_EXEC_PATH=${PATH}::
export OCTAVE_PATH=${STRUCTEDIT_PATH}::

ulimit -s unlimited
alias octave="octave -p $OCTAVE_PATH"
# -----
Do you want to continue (Y/n)? Y
```

```
...
Edit .rhosts file now? (y/N) N
...
[username@localhost WIEN2k]$ source ~/.bashrc
...
```

11. To setup or start w2web, enter in the terminal:

```
[username@localhost WIEN2k]$ w2web
#####
# w2web starter                                     #
# Copyright (C) 2001 luitz.at                       #
#####
w2web installer on host localhost

#####
# w2web installer                                     #
# Copyright (C) 2001 luitz.at                       #
#####
```

Checking for Installation in /home/username/.w2web/localhost

Creating /home/username/.w2web
Creating /home/username/.w2web/localhost
conf directory does not exist - creating it.
logs directory does not exist - creating it.
sessions directory does not exist - creating it.
tmp directory does not exist - creating it.

Installing w2web files ...
Please answer these questions for proper installation.
Just press enter for the default value of (in brackets).

Please enter the username: [admin] username
Please enter the password: [password] password
username:password
Remember these. You will need them when you log in.

Select the port to run on: [7890]
Running on port 7890

Please enter this system's hostname: [localhost] localhost
Using localhost

Is this your master node?: [y] y

Installing... Attempting to start now...

Trying to start /home/username/WIEN2k/SRC_w2web/bin/w2web
w2web server started, now point your web browser to
<http://localhost:7890>

done.

12. The WIEN2k 24.1 installation is now complete. To test if WIEN2k installed okay or not, run the TiC example. There is currently not a specific TiC file for Rocky Linux, but you may try following instead the Fedora file at [1]. Additional details of the TiC example are given in the WIEN2k usersguide (section “3 Quick Start” starting on page 13) [2].

[1] [https://github.com/gsabo/WIEN2k-Docs/blob/main/WIEN2k24.1%20TiC%20Example\(Fedora-ifx\).pdf](https://github.com/gsabo/WIEN2k-Docs/blob/main/WIEN2k24.1%20TiC%20Example(Fedora-ifx).pdf)

[2] http://susi.theochem.tuwien.ac.at/reg_user/textbooks/usersguide.pdf